

Language Sensitivity Is Not Directional Control

Matched static-instruction interventions in robot policies and world-action models

Across eight expanded checkpoints, changing the static instruction changed every matched action trace. Endpoint redirection was frequent, but requested task completion remained checkpoint- and direction-dependent.

PRIMARY EVIDENCE: 648 PHASE-A EPISODES | SEPARATE 40-EPISODE COMPATIBILITY COHORT | 108-EPISODE ABLATION

DROID/RoboLab and RoboTwin use different tasks, controllers, and success predicates. Their success rates are reported separately and never pooled.

Question, hypothesis, and answer

The intervention changes one semantic relation while preserving the matched reset and runtime identity.

Question. From an identical scene and seed, does changing only the static instruction from LEFT to RIGHT (1) change the executed actions, (2) order the final endpoints accordingly, and (3) preserve task competence in both directions?

Hypothesis. A controllable policy should be sensitive to the instruction, redirect the object in the requested ordering, and complete the frozen relation-and-release task under both commands.

Answer

What generalized

All 135 DROID Phase-A pairs and all 189 RoboTwin replicate pairs produced distinct executed action traces. Endpoint ordering aligned in 125/135 DROID pairs and 128/189 RoboTwin pairs.

What did not

Task completion depended strongly on checkpoint and requested direction. GR00T changed all action traces but completed only 3/54 episodes; Nano completed 51/54 while still exhibiting geometry-sensitive side depth.

Scientific reading. Language sensitivity is necessary evidence of conditioning, not sufficient evidence of directional control. A model can react to the word, move in the requested ordering, and still fail to pick, transport, release, or robustly satisfy both directions.

The exact intervention

LEFT and RIGHT are shorthand labels for full static prompts, not isolated tokens.

DROID / ROBOLAB - LEFT CONDITION

Put the Rubik's cube to the left of the bowl.

DROID / ROBOLAB - RIGHT CONDITION

Put the Rubik's cube to the right of the bowl.

RoboTwin uses the same sentence structure with scene-specific objects. For example, pair03 changes only left to right in: "Put the small woodenblock to the [left/right] of the red playingcards box." The seven expanded scenes use pair03-pair09; all full strings appear in Appendix B.

Matched invariants

Scene and reset

same initial state hash within each pair

Seeds

same environment and sampling seed

Runtime

same checkpoint, adapter, controller, and horizon

Instruction

static for the full episode; relation word changes

Scoring

frozen arena-specific predicate and failure taxonomy

Evidence

video and raw per-episode record for every valid rollout

No oracle, subtask coach, prompt switching, or progress-conditioned language. Every valid behavioral failure stays in the denominator; infrastructure-invalid and partial attempts remain separate.

Three estimands, kept separate

The analysis asks whether language changes behavior, whether the change has the requested sign, and whether the task is completed.

1

Action sensitivity

LEFT and RIGHT executed action arrays differ over their common prefix. This detects conditioning, not correctness.

2

Endpoint redirection

The matched RIGHT endpoint is ordered to the requested side of the LEFT endpoint. This measures semantic direction.

3

Task competence

The rollout satisfies the frozen requested relation and detached-release predicate. This is the success denominator.

Continuous fields retained for every episode

Signed final lateral offset; requested-side margin; cone-entry time; episode length; sustained versus transient entry; time to first contact where the adapter exposes it; and object path length. Margin is success-conditional when used as a competence measure, whereas signed offset remains defined on failures.

Evidence accounting

648

completed Phase-A episodes

270 DROID + 378 RoboTwin

40

separate pi0-FAST compatibility episodes

20 matched pairs; not historical recovery

108

Nano position-reflection episodes

27 seeds x 2 layouts x 2 prompts

160

historical wording episodes

two checkpoints; separate V1 cohort

Sensitivity was widespread; control was not

Every expanded checkpoint changed its executed trace, but endpoint alignment and success remained checkpoint-specific.

DROID / ROBOLAB - 27 MATCHED PAIRS PER CHECKPOINT

pi0.5 current stack

VLA | 54 episodes

Action response 27/27 Endpoint aligned 25/27 LEFT 5/27 RIGHT 24/27

GR00T N1.7

VLA | 54 episodes

Action response 27/27 Endpoint aligned 21/27 LEFT 3/27 RIGHT 0/27

Cosmos3 Edge Policy

WAM | 54 episodes

Action response 27/27 Endpoint aligned 27/27 LEFT 18/27 RIGHT 25/27

Cosmos3 Nano Policy

WAM | 54 episodes

Action response 27/27 Endpoint aligned 27/27 LEFT 26/27 RIGHT 25/27

DreamZero action guidance s=2

WAM | 54 episodes

Action response 27/27 Endpoint aligned 25/27 LEFT 3/27 RIGHT 17/27

ROBOTWIN - 63 REPLICATE PAIRS PER CHECKPOINT, NESTED WITHIN SEVEN SCENES

Efficient-WAM-RT

WAM | 126 episodes

Action response 63/63 Endpoint aligned 42/63 LEFT 26/63 RIGHT 28/63

FastWAM

WAM | 126 episodes

Action response 63/63 Endpoint aligned 39/63 LEFT 24/63 RIGHT 20/63

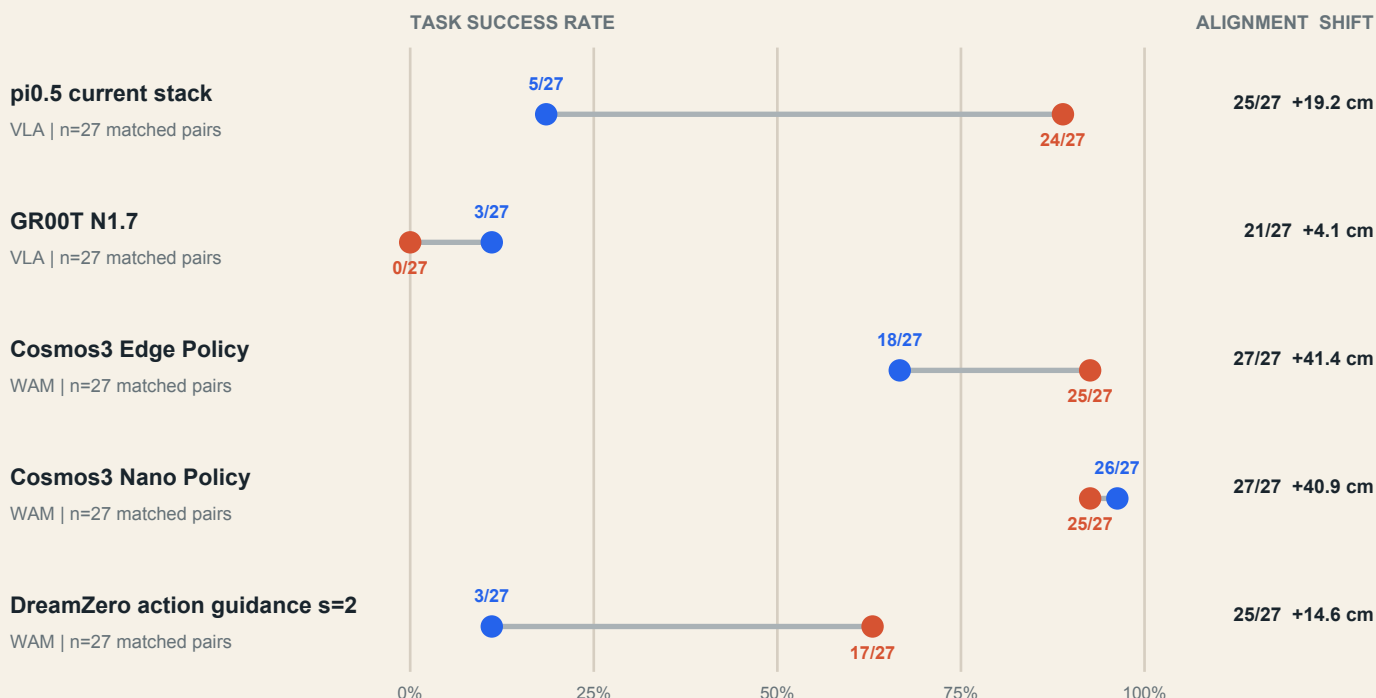
LingBot-VA

WAM | 126 episodes

Action response 63/63 Endpoint aligned 47/63 LEFT 19/63 RIGHT 19/63

DROID: direction-dependent competence

Task success, endpoint ordering, and mean requested-oriented shift are shown on the same checkpoint row.



Reading the pattern. Nano remained near ceiling in both directions (26/27 LEFT; 25/27 RIGHT), while pi0.5 and DreamZero favored RIGHT and GR00T favored LEFT weakly but failed almost everywhere. Edge completed both directions more often, yet RIGHT still exceeded LEFT by 7/27 episodes. High alignment therefore does not erase checkpoint-specific directional asymmetry.

Separate public compatibility cohort

pi0-FAST public compatibility

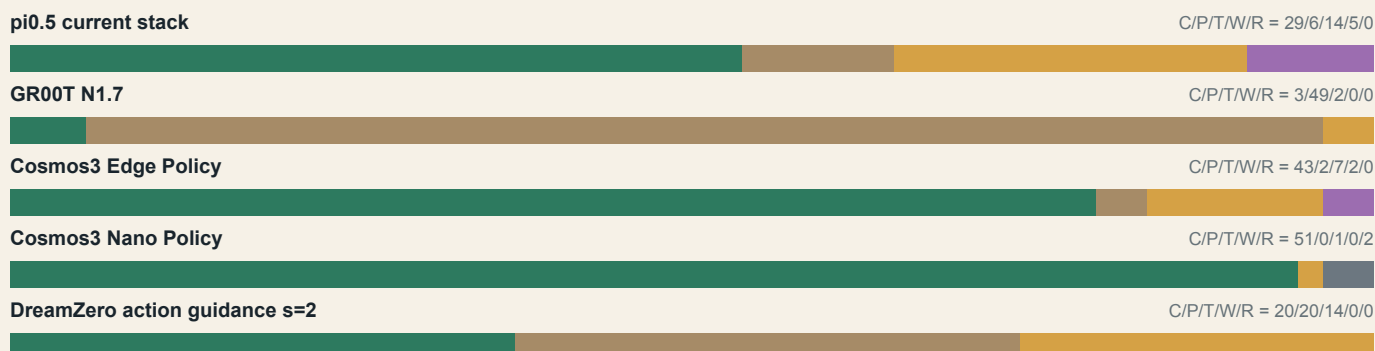
LEFT 0/20

RIGHT 12/20

ALIGNED 16/20

All 20 action prefixes differed; exact McNemar $p=0.000488$ and endpoint sign-test $p=0.00443$. This is a public old-name-config bridge, not recovery of the unavailable historical integration.

Failure decomposition



RoboTwin: response across seven scenes

Nine sampling replicates are nested within each of seven scenes; 63 replicate pairs are not 63 independent scenes.

Prompt sensitivity is common; directional control and task completion are not

RoboTwin expanded matched-pair experiment · seven scenes × nine new replicates · 63 pairs / 126 episodes per checkpoint · valid failures retained

Efficient-WAM-RT

decoded future video + actions

LEFT prompt

task success · episode unit

26 / 63 · 41%

RIGHT prompt

task success · episode unit

28 / 63 · 44%

Endpoint ordering

requested order · pair unit

42 / 63 · 67%

Action traces

first 10 actions · pair unit

63 / 63 · 100%

FastWAM

action-only at test time

LEFT prompt

task success · episode unit

24 / 63 · 38%

RIGHT prompt

task success · episode unit

20 / 63 · 32%

Endpoint ordering

requested order · pair unit

39 / 63 · 62%

Action traces

first 10 actions · pair unit

63 / 63 · 100%

LingBot-VA

latent future + actions

LEFT prompt

task success · episode unit

19 / 63 · 30%

RIGHT prompt

task success · episode unit

19 / 63 · 30%

Endpoint ordering

requested order · pair unit

47 / 63 · 75%

Action traces

first 10 actions · pair unit

63 / 63 · 100%

Every pair changed

189 / 189

distinct action traces

Direction was respected

128 / 189

endpoint order aligned

Full task completed

136 / 378

valid episodes

What was consistent

All 189 replicate pairs changed their first ten executed actions. Endpoint ordering aligned in 42/63 Efficient-WAM-RT, 39/63 FastWAM, and 47/63 LingBot-VA pairs.

What remained model- and scene-dependent

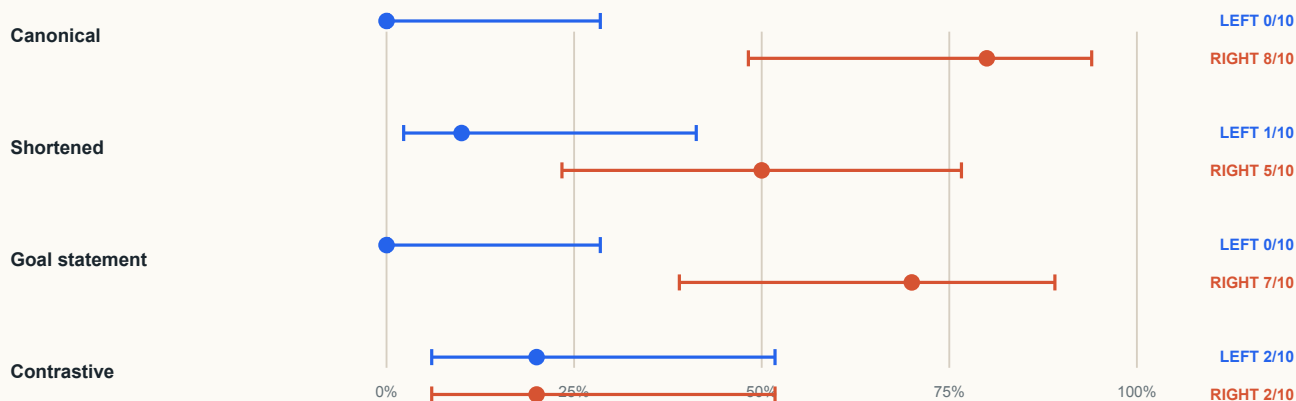
Requested success was 54/126, 44/126, and 38/126 respectively. These conditional replicate counts are descriptive; arena success is never pooled with DROID.

Phrasing interacted with requested direction

A separate 160-episode V1 DROID cohort tested four static prompt forms at ten matched seeds per cell.

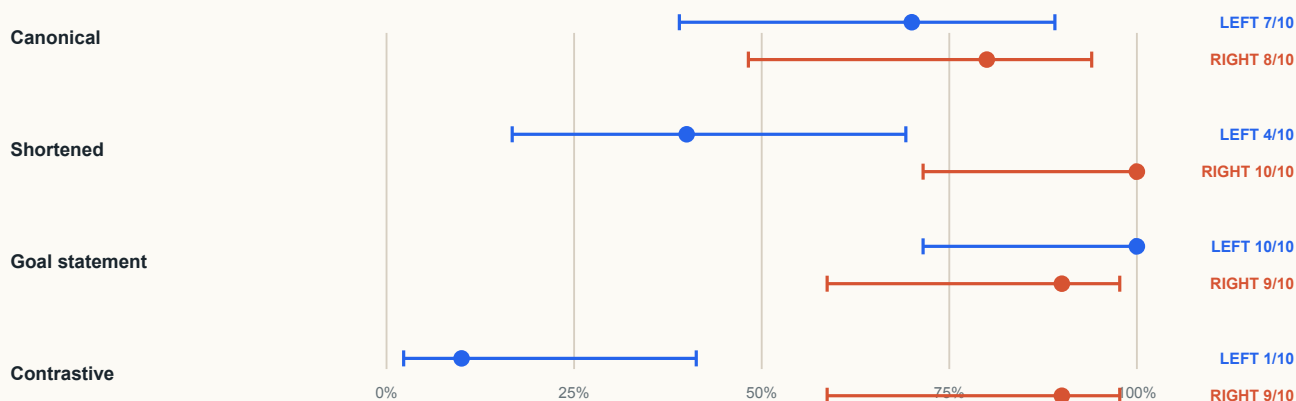
pi0.5 DROID

Task-success rate with Beta(1,1) posterior 95% credible interval



Cosmos3 Edge DROID

Task-success rate with Beta(1,1) posterior 95% credible interval



The result is an interaction, not a checkpoint-wide score. pi0.5 ranged from 0/10 LEFT versus 8/10 RIGHT under the canonical wording to 2/10 versus 2/10 under the contrastive wording. Cosmos3 Edge ranged from 10/10 versus 9/10 under the goal statement to 1/10 versus 9/10 under the contrastive wording. The contrastive Cosmos direction gap had exact paired $p=0.00781$; the exploratory wording-pair tests were not multiplicity corrected.

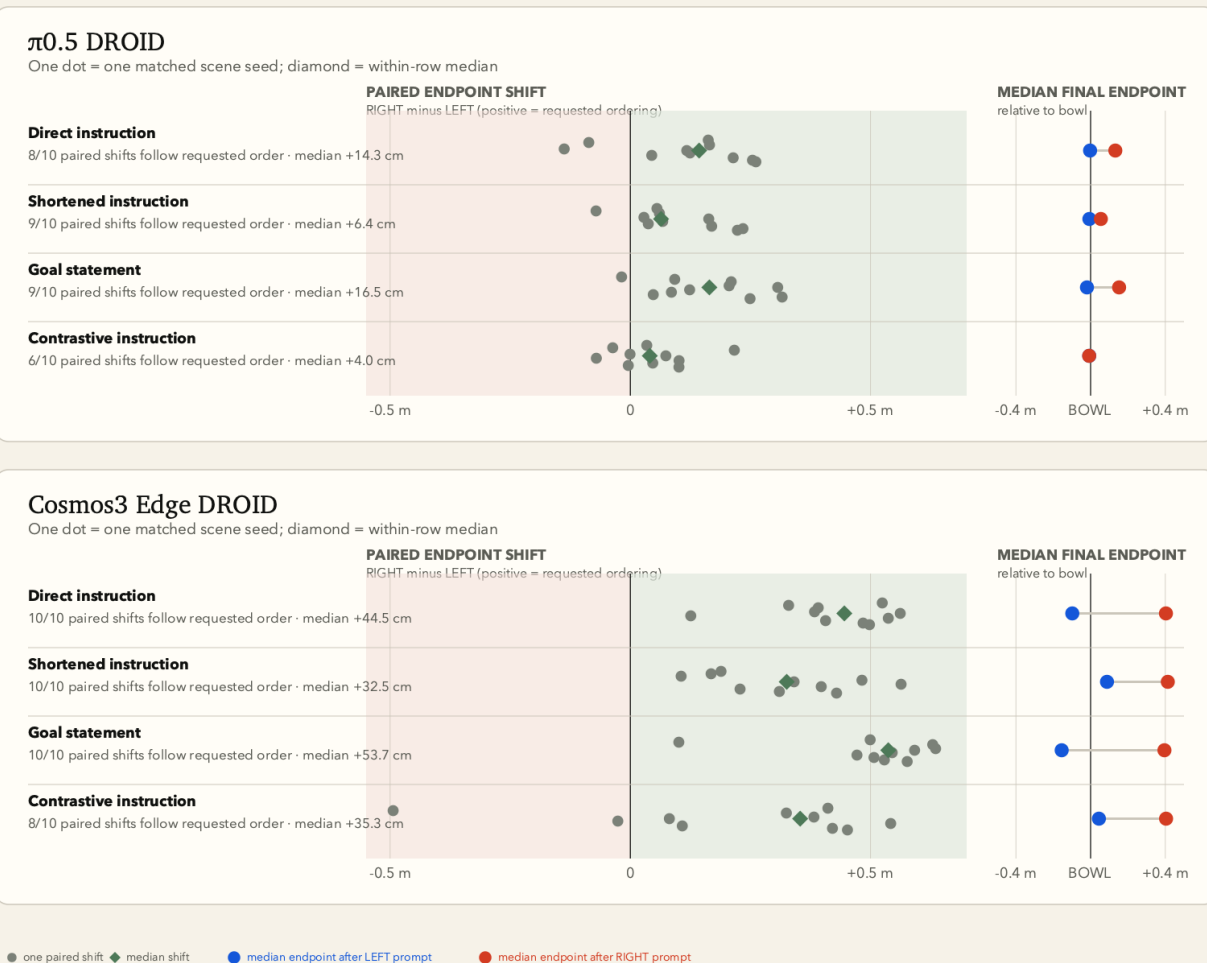
These historical V1 prompts were lower-case and unpunctuated; Appendix B prints the byte-level strings. This cohort is not the unreleased 480-episode V3 Phase C grid.

Physical redirection exceeded reliable completion

Each dot is one matched RIGHT-minus-LEFT endpoint shift; the plot measures redirection, not competence.

Matched prompts redirect endpoints more often than they complete the task

DROID wording experiment · matched scene seeds · endpoint shift is a physical-response diagnostic, not a task-success score



$\pi 0.5$ endpoints moved in the requested LEFT-to-RIGHT ordering in 32/40 wording-matched pairs (median shift +9.7 cm); Cosmos3 Edge did so in 38/40 (median +41.6 cm). Endpoint motion can therefore be prompt-conditioned even when pick, transport, release, or the full task predicate fails.

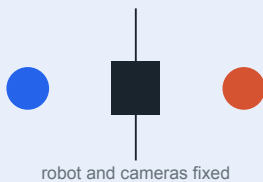
The right-hand mini-panels locate median LEFT- and RIGHT-condition endpoints relative to the bowl. Success remains the full relation-and-release predicate, not lateral position alone.

Nano position-reflection ablation

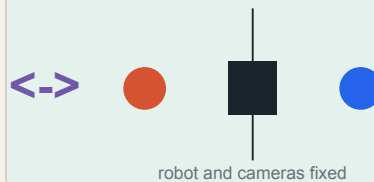
The registered intervention reflected movable-object center positions while holding robot, cameras, and nonmovable geometry fixed.

Question. Does Nano's directional margin asymmetry change when only movable-object center positions are reflected about the robot sagittal plane?

CONTROL LAYOUT



POSITION-REFLECTED LAYOUT



27 prespecified seeds x 2 layouts x 2 static prompts = 108/108 valid episodes. Success was control 26/27 LEFT and 26/27 RIGHT; reflected 27/27 LEFT and 23/27 RIGHT. The intervention is positions-only, not a full-scene mirror.

Two geometric estimands

Endpoint redirection D

RIGHT-condition final endpoint minus LEFT-condition final endpoint, oriented so positive means the requested LEFT-to-RIGHT ordering.

Requested-depth contrast B

RIGHT requested-side margin minus LEFT requested-side margin. Positive means deeper placement under RIGHT than LEFT; negative reverses that contrast.

Primary interaction tests. J asks whether position reflection changes endpoint redirection. I asks whether it changes requested-depth contrast. Bootstrap resampling and exact sign tests use the matched seed as the unit.

Redirection persisted; side depth reversed

All seed-level points are shown. Means use matched-seed bootstrap 95% intervals; p-values are exact two-sided sign tests.

Endpoint redirection

The prompt still ordered endpoints in all 27 seeds under both layouts.

Control D: mean +44.8 cm [40.2, 49.4]



Reflected D: mean +45.3 cm [40.5, 50.3]



Interaction J: mean +0.5 cm [-5.6, +6.6], p=0.701



Requested-side depth

The same prompt redirection persisted, but how deeply the model placed the object on each requested side changed with geometry.

Control B: mean +14.8 cm [+9.6, +19.7]



Reflected B: mean -10.0 cm [-14.8, -5.3]



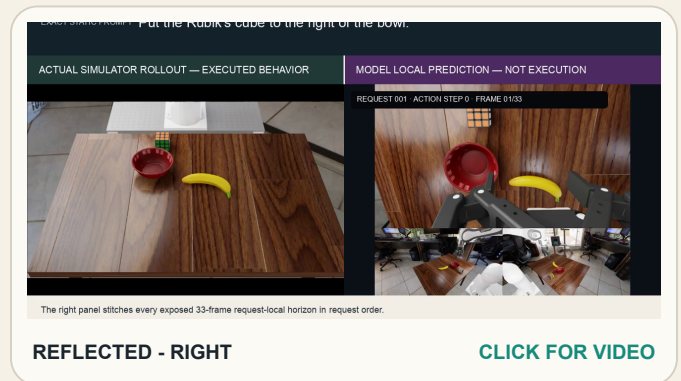
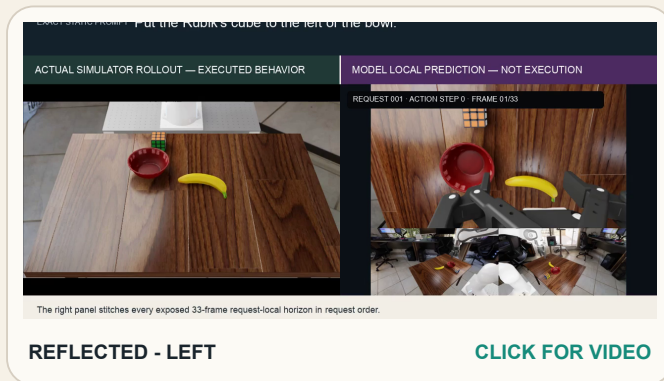
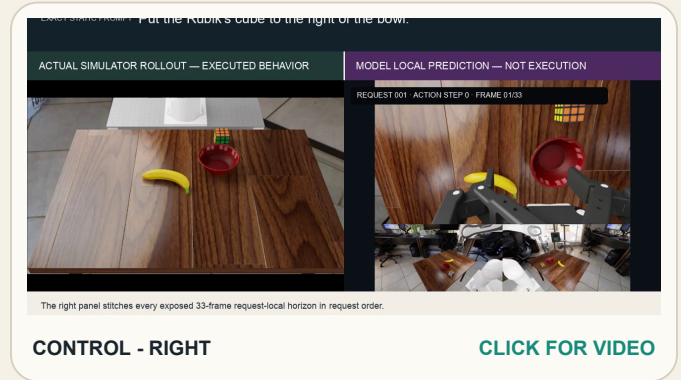
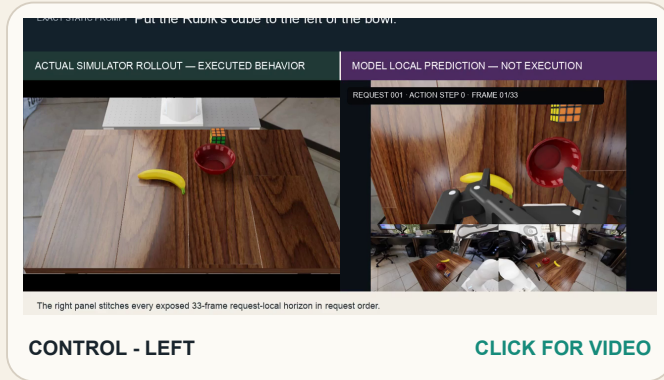
Interaction I: mean -24.8 cm [-32.4, -17.3], p=4.92e-05



Interpretation. The model-side language response remained strong, but the quantitative quality of the requested placement was causally altered by the registered object-position intervention. This does not identify training distribution, reachability, or embodiment as the mechanism.

Failures and future-interface evidence

Actual simulator execution and request-local predictions are displayed together but remain analytically distinct.



Prediction boundary. The right-hand video panels concatenate request-local 33-frame predictions returned during control. They are not one continuous full-task imagination, not robot execution, and not additional episodes.

Failure taxonomy: 108 episodes

102 correct | 5 transport_failed | 1 release_failed | 0 pick_failed | 0 wrong_side

The cohort retained 738 decoded local predictions and 21,972 executed actions. Success-conditional interaction analysis uses only the 21 seeds correct in all four cells; no failure is converted to zero.

World-model evidence is not robot execution

Decoded videos, latent futures, and action-only policies expose different evidence and must not share a success denominator.

Checkpoint	Executed rollouts	Future evidence	How it is used
Cosmos3 Edge Policy	54	452 decoded local futures	interface evidence; not extra trials
Cosmos3 Nano Policy	54	349 local predictions	request-local horizons; not full-task imagination
DreamZero s=2	54	54 official full-reset decodes + 2,554 latent futures	imagination reported beside, not as, execution
Efficient-WAM-RT	126	126 decoded coarse futures	RoboTwin future evidence
FastWAM	126	action-only at test time	no missing future encoded as zero
LingBot-VA	126	126 latent-only futures	latent-only, never substituted with video
pi0.5 / GR00T / pi0-FAST	54 / 54 / 40	action-only	VLA execution evidence

Small guidance ablation (descriptive)

Nano local guidance g=3 to g=1

Success 6/6 to 4/6. LEFT 3/3 to 1/3; RIGHT 3/3 to 3/3. The requested-margin gap shrank because RIGHT depth fell more; weak LEFT placement did not improve.

DreamZero s=1 to derived action guidance s=2

Success 3/6 to 4/6. LEFT 2/3 to 1/3; RIGHT 1/3 to 3/3. Bias moved toward RIGHT; this is redistribution, not a powered improvement claim.

Both guidance comparisons use only three seeds per direction and remain descriptive. DreamZero s=2 is a derived negative-branch action-guidance intervention, not official action-CFG. Additional Cosmos3 Super 64B and Cosmos3 Edge base artifacts are prediction/action probes without simulator execution; Cosmos-Reason2 has no behavioral denominator.

What the evidence supports

The strongest claim is checkpoint-specific: language reliably perturbs behavior, but semantic control and competence can separate.

Supported

- Within an exact checkpoint and matched reset, changing the static relation word caused different executed action traces.
- Endpoint ordering often followed the requested direction, including in many task failures.
- Task competence remained checkpoint-, direction-, wording-, and scene-dependent.
- Nano's registered positions-only intervention causally changed requested-side depth while preserving endpoint redirection.

Not supported

- A universal VLA-versus-WAM ranking or a pooled cross-arena success rate.
- Training-data imbalance as the mechanism behind directional asymmetry.
- Isolation of reachability, embodiment, base handedness, or a full-scene mirror.
- Decoded or latent futures as additional trials, robot outcomes, or proof that imagination caused execution.

A more precise story than "the model succeeded"

Success is a binary endpoint of a longer causal chain: recognize the prompt, select a distinct action sequence, contact and pick the object, transport it in the requested semantic direction, enter and sustain the target region, and release. The expanded data show failures at different links. Directional asymmetry can therefore coexist with both high sensitivity and apparently plausible trajectories.

The practical implication is that best-of-N or language-guided sampling should optimize a continuous quality measure such as requested-side margin, while retaining the full task predicate and failure taxonomy. Binary success alone is insensitive at ceiling and opaque about why a rollout failed.

Limitations and current status

The report distinguishes completed evidence from registered or externally blocked work.

Checkpoint and simulator scope

Results identify exact runtimes, controllers, action spaces, and scene distributions. They do not establish a model-family ranking.

Arena separation

DROID/RoboLab and RoboTwin use different tasks and success rules. Their rates are never pooled.

RoboTwin nesting

Nine stochastic replicates are nested within each of seven scenes. Episode-level intervals do not represent 63-scene generalization.

Historical pio-FAST

Exact historical OpenPI/RoboLab revisions remain unavailable. The 40-episode public compatibility cohort is separate and cannot replace them.

Measurement coverage

The committed 982/982 audit covers pre-B001 episodes; the later 108 B001 episodes are separately complete. No new 1,090-episode pooled audit is claimed.

Future interfaces

Decoded, local, full-reset, latent-only, and action-only evidence are labeled separately; missing futures are never zeros.

Released and unreleased work

Layer	Status	Accounting
V3 Phase A	complete	648/648 authorized-new episodes
pi0-FAST V3-A002	complete, separate	40 episodes; not historical recovery
Nano V3-B001	complete	108/108 valid episodes
Other Phase B ablations	unreleased	no result claimed
V3 Phase C wording	registered, unreleased	0/480 released episodes
Eligible Phase D repeats	unreleased	no result claimed

Appendix A: DROID checkpoint table

Expanded Phase A and the public compatibility cohort are separated from bounded historical reference evidence.

Cohort / checkpoint	Type	Episodes	LEFT	RIGHT	Aligned	Actions	Shift	C/P/T/W/R	Future interface
pi0.5 current stack	VLA	54	5/27	24/27	25/27	27/27	+19.2 cm	29/6/14/5/0	action-only
GR00T N1.7	VLA	54	3/27	0/27	21/27	27/27	+4.1 cm	3/49/2/0/0	action-only
Cosmos3 Edge Policy	WAM	54	18/27	25/27	27/27	27/27	+41.4 cm	43/2/7/2/0	452 decoded local futures
Cosmos3 Nano Policy	WAM	54	26/27	25/27	27/27	27/27	+40.9 cm	51/0/1/0/2	349 decoded local futures
DreamZero action guidance s=2	WAM	54	3/27	17/27	25/27	27/27	+14.6 cm	20/20/14/0/0	54 full-reset decodes + 2,554 latent futures
SEPARATE: pi0-FAST public compatibility	VLA	40	0/20	12/20	16/20	20/20	+16.6 cm	12/22/4/0/2	action-only
BOUNDED V2: pi0-FAST frozen reference	VLA	20	1/10	10/10	10/10	10/10	+27.9 cm	not harmonized	action-only

How to read the table. Shift is the requested-oriented mean endpoint displacement: positive follows the requested LEFT-to-RIGHT ordering. C/P/T/W/R means correct, pick_failed, transport_failed, wrong_side, release_failed. The bounded historical row predates the harmonized V3 taxonomy and is not combined with the compatibility cohort.

All five expanded DROID checkpoints have 27/27 distinct action pairs. Wilson intervals and exact paired tests are retained in the source summaries; the main text reports the named tests and avoids cross-arena pooling.

Appendix A: RoboTwin checkpoint table

Expanded replicate evidence and bounded references are shown in separate rows; no DROID result is pooled here.

Cohort / checkpoint	Type	Episodes	LEFT	RIGHT	Aligned	Actions	Mean / median shift	C/P/T/W/R	Future interface
Efficient-WAM-RT	WAM	126	26/63	28/63	42/63	63/63	-14.4 / +4.4 cm	54/28/38/6/0	126 decoded coarse futures
FastWAM	WAM	126	24/63	20/63	39/63	63/63	-0.4 / +1.8 cm	44/31/36/15/0	action-only at test time
LingBot-VA	WAM	126	19/63	19/63	47/63	63/63	+19.9 / +11.0 cm	38/9/64/15/0	126 latent-only futures
BOUNDED V2: LingBot-VLA 4B	VLA	6	1/3	0/3	2/3	not reported	-0.9 cm	not harmonized	none
BOUNDED V2: Light-WAM	WAM	6	1/3	0/3	1/3	3/3	-3.5 cm	not harmonized	action-only

Nesting and outliers. The 63 expanded pairs per checkpoint are nine stochastic replicates nested within seven scenes. Efficient-WAM-RT has rare reverse-tail outliers that flip its mean shift negative (-14.4 cm) even though the median remains positive (+4.4 cm); report both. FastWAM is near zero by both summaries, while LingBot-VA shows a positive mean and median.

Historical r00 coverage remains separate: Efficient-WAM-RT 3/7 LEFT and 2/7 RIGHT; FastWAM 1/7 and 1/7; LingBot-VA 3/7 and 4/7. Those rows are not added to the expanded denominators above.

Appendix B: exact DROID prompts

Every LEFT and RIGHT string is printed in full. Historical V1 wording is not byte-identical to V3.

Current V3 direct command

LEFT Put the Rubik's cube to the left of the bowl.

RIGHT Put the Rubik's cube to the right of the bowl.

Historical V1 wording layer

CANONICAL

LEFT Put the rubiks cube to the left of the bowl

RIGHT Put the rubiks cube to the right of the bowl

SHORT

LEFT Put the cube left of the bowl

RIGHT Put the cube right of the bowl

DECLARATIVE

LEFT The rubiks cube should end up to the left of the bowl

RIGHT The rubiks cube should end up to the right of the bowl

CONTRASTIVE

LEFT Put the rubiks cube to the left of the bowl, not to the right of the bowl

RIGHT Put the rubiks cube to the right of the bowl, not to the left of the bowl

Cohort boundary. These eight V1 strings were lower-case and unpunctuated. They produced the completed 160-episode historical wording layer. They are not the title-cased, punctuated prompts registered for unreleased V3 Phase C.

Appendix B: exact RoboTwin prompts

pair03-pair09 are the expanded V3 scene set; pair00-pair02 are bounded historical scenes.

PAIR00 BOUNDED

LEFT Put the blue soap to the left of the tea-box.

RIGHT Put the blue soap to the right of the tea-box.

PAIR01 BOUNDED

LEFT Put the box of playingcards to the left of the rubikscube.

RIGHT Put the box of playingcards to the right of the rubikscube.

PAIR02 BOUNDED

LEFT Put the box with cards inside to the left of the red coffee-box.

RIGHT Put the box with cards inside to the right of the red coffee-box.

PAIR03 EXPANDED

LEFT Put the small woodenblock to the left of the red playingcards box.

RIGHT Put the small woodenblock to the right of the red playingcards box.

PAIR04 EXPANDED

LEFT Put the plastic mouse to the left of the blue stapler.

RIGHT Put the plastic mouse to the right of the blue stapler.

PAIR05 EXPANDED

LEFT Put the box of playingcards to the left of the rubikscube.

RIGHT Put the box of playingcards to the right of the rubikscube.

PAIR06 EXPANDED

LEFT Put the coffee box to the left of the red playingcards box.

RIGHT Put the coffee box to the right of the red playingcards box.

PAIR07 EXPANDED

LEFT Put the golden bread to the left of the blue stapler.

RIGHT Put the golden bread to the right of the blue stapler.

PAIR08 EXPANDED

LEFT Put the box with cards inside to the left of the black phone.

RIGHT Put the box with cards inside to the right of the black phone.

PAIR09 EXPANDED

LEFT Put the rubikscube to the left of the brown woodenblock.

RIGHT Put the rubikscube to the right of the brown woodenblock.

Each matched pair holds scene, reset, seeds, runtime, controller, and horizon fixed. Only the complete static instruction's requested relation changes.

Appendix C: publication media index

PDF buttons open committed repository videos. Actual execution, model prediction, and imagination remain labeled separately.

DROID / RoboLab prompt pair

LEFT Put the Rubik's cube to the left of the bowl. **RIGHT** Put the Rubik's cube to the right of the bowl.

DROID execution and future media

pi0-FAST compatibility

ACTUAL

pi0.5 current stack

ACTUAL

GR00T N1.7

ACTUAL

Cosmos3 Edge Policy

ACTUAL

Cosmos3 Nano Policy

ACTUAL

Cosmos3 Nano Policy

PREDICTION

DreamZero

ACTUAL

DreamZero

FULL IMAGINATION

RoboTwin execution media

Efficient-WAM-RT - pair05

LEFT Put the box of playingcards to the left of the rubikscube.

RIGHT Put the box of playingcards to the right of the rubikscube.

ACTUAL

FastWAM - pair02

LEFT Put the box with cards inside to the left of the red coffee-box.

RIGHT Put the box with cards inside to the right of the red coffee-box.

ACTUAL

LingBot-VA - pair03

LEFT Put the small woodenblock to the left of the red playingcards box.

RIGHT Put the small woodenblock to the right of the red playingcards box.

ACTUAL

LingBot-VLA 4B - pair00

LEFT Put the blue soap to the left of the tea-box.

RIGHT Put the blue soap to the right of the tea-box.

ACTUAL

Light-WAM - pair00

LEFT Put the blue soap to the left of the tea-box.

RIGHT Put the blue soap to the right of the tea-box.

ACTUAL

Prediction-only probes. Cosmos3 Super 64B and Cosmos3 Edge base have committed prediction/action videos but no robot-execution score. Cosmos-Reason2 has no behavioral video and is not represented by substituted media.

EDGE BASE

SUPER PROBE

Appendix D: methods and provenance

The PDF is generated from committed summaries and selected publication assets; raw rollouts and checkpoints remain outside ordinary Git.

Metric and statistical definitions

Requested success

Frozen arena-specific relation-and-detached-release predicate. All valid behavioral failures remain in the denominator.

Endpoint alignment

Strict requested ordering of the matched LEFT and RIGHT final lateral endpoints. Exact sign tests exclude ties.

Requested-side margin

Signed geometric depth into the requested region. Success-conditional analyses omit failures; no missing value becomes zero.

Intervals

V3 direction rates use Wilson 95% intervals. Nano means use 20,000 matched-seed bootstrap replicates. V1 wording intervals are Beta(1,1) posterior credible intervals.

Paired tests

Exact two-sided McNemar tests use discordant success pairs. Exact two-sided sign tests use non-tied geometric differences.

Failure taxonomy

Precedence: correct; pick_failed; wrong_side; release_failed; transport_failed. Original frozen stage remains retained.

Authoritative evidence map

- docs/VLA_WAM_STEERABILITY_V3_PROTOCOL.md
- docs/VLA_WAM_V3_CONTINUATION.md
- artifacts/vla_wam_shared_v3/continuation_state.json
- artifacts/vla_wam_shared_v3/results/*_phase_a_summary.json
- artifacts/vla_wam_shared_v3/results/pi0_fast_old_name_config_v3a002_summary.json
- artifacts/vla_wam_shared_v3/phase_b/nano_mirror_v3b001/results/nano_v3b001_summary.json
- artifacts/vla_wam_shared_v3/measurement_coverage_audit.json
- artifacts/vla_wam_shared_v1/run_manifest.json and final_evidence/

Rebuild contract. Run `tmp/pdfs/.venv/bin/python tools/build_vla_wam_research_pdf.py`. The builder reads the committed JSON summaries above, selected publication figures under `artifacts/vla_wam_shared_v3/publication_pdf/`, and committed Nano posters. It writes a deterministic PDF and SHA-bearing manifest under `output/pdf/`.